

# Results Analysis

Influence of Sedimentary Segregation on the Dynamics of Hydraulic  
Dunes: Bidisperse Model

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## Contents

|          |                                                                                |           |
|----------|--------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction and Model Context</b>                                          | <b>2</b>  |
| <b>2</b> | <b>Compositional Fan: Space–Time Overview of Segregation</b>                   | <b>3</b>  |
| <b>3</b> | <b>Averaged Vertical Sorting Profile</b>                                       | <b>6</b>  |
| <b>4</b> | <b>Velocity Field in the Co-moving Frame and Trapped Coarse Zone</b>           | <b>8</b>  |
| <b>5</b> | <b>Controlled Comparison: Effect of Granular Diffusion</b>                     | <b>10</b> |
| <b>6</b> | <b>Temporal Evolution of <math>\phi_s</math>: Kinetics of Vertical Sorting</b> | <b>16</b> |
| <b>7</b> | <b>Equilibrium Length <math>\ell</math> and Local Péclet Fields</b>            | <b>19</b> |
| <b>8</b> | <b>Summary of the Main Results</b>                                             | <b>21</b> |
| <b>A</b> | <b>Temporal Convergence Analysis</b>                                           | <b>23</b> |

# 1 Introduction and Model Context

This document analyzes the figures generated by the numerical model of sedimentary segregation in bidisperse hydraulic dunes. The system studies how mixtures of two grain sizes —fines ( $d_s = 0.30$  mm) and coarse ( $d_l = 1.00$  mm)— reorganize vertically while the dune migrates under an oscillatory hydraulic flow.

The narrative is organized to build physical understanding progressively: we begin with the global space–time picture of the segregation process (Sec. 2), then zoom into the time-averaged vertical structure (Sec. 3), reveal the flow mechanism responsible for trapping coarse grains (Sec. 4), isolate the role of granular diffusion (Sec. 5), examine the time evolution of the profile (Sec. 6), and finally quantify the length-scales that govern the segregation–diffusion competition (Sec. 7). Temporal convergence of the simulations is documented in Appendix A.

## Fundamental variables and conventions

- $\phi_s$  is the **volumetric fraction of fines**:  $\phi_s = 1$  corresponds to pure fine sediment and  $\phi_s = 0$  to pure coarse.
- $\eta = z/h$  is the **normalized vertical coordinate**:  $\eta = 0$  is the impermeable base and  $\eta = 1$  the free surface. The local bed thickness varies between  $h = 1.00$  mm (flat bed) and  $h = 9.00$  mm (crest).
- $\langle \phi_s \rangle_x(\eta)$  is the **average in  $x$**  over the dune body, weighted by  $h(x)$  (volume average).
- $\eta_a$  is the **roof of the buried deposit**: below this height the material has already left the active layer ( $\delta_a = 1.5$  mm) and its composition is frozen ( $\langle \eta_a \rangle \approx 0.71$ ).
- $\ell$  is the **segregation–diffusion equilibrium length**:

$$\ell = \frac{D_{sl}}{f_{sl}} = \frac{A(C\bar{d} + p)}{B F(R, \phi_s)} \quad (1)$$

where  $\dot{\gamma}$  cancels between  $f_{sl}$  and  $D_{sl}$ , so  $\ell$  does not depend on the hydraulic slope  $i$ .

- The **local Péclet number**  $Pe = L/\ell$  measures how many interface thicknesses fit in the reference scale  $L$ :
  - $L = h$ : scale of the entire column.
  - $L = \delta_a$ : scale of the active layer (physically relevant scale for burial).

The main runs compare two physical configurations:

1. **With granular diffusion** ( $A = 0.108$ , Trehwela et al. 2021): the diffusive term  $D_{sl}$  broadens the compositional interfaces.
2. **Without diffusion** (pure hyperbolic model,  $D_{sl} = 0$ ): upper bound of the compositional contrast; the interfaces only broaden due to numerical dissipation.

All runs in the main section were carried out until  $t = 2400$  s, with mesh  $N_x = 400$ ,  $N_z = 40$ ,  $U_0 = 3 \times 10^{-4}$  m/s, initial concentration  $\phi_s^0 = 0.70$  unless otherwise indicated.

## 2 Compositional Fan: Space–Time Overview of Segregation

The compositional fan is the starting point of the analysis because it provides the most complete picture of the system: the entire temporal history of  $\phi_s(x, \eta)$  is visible at once, revealing *how quickly* and *where* segregation reorganizes the sediment mixture.

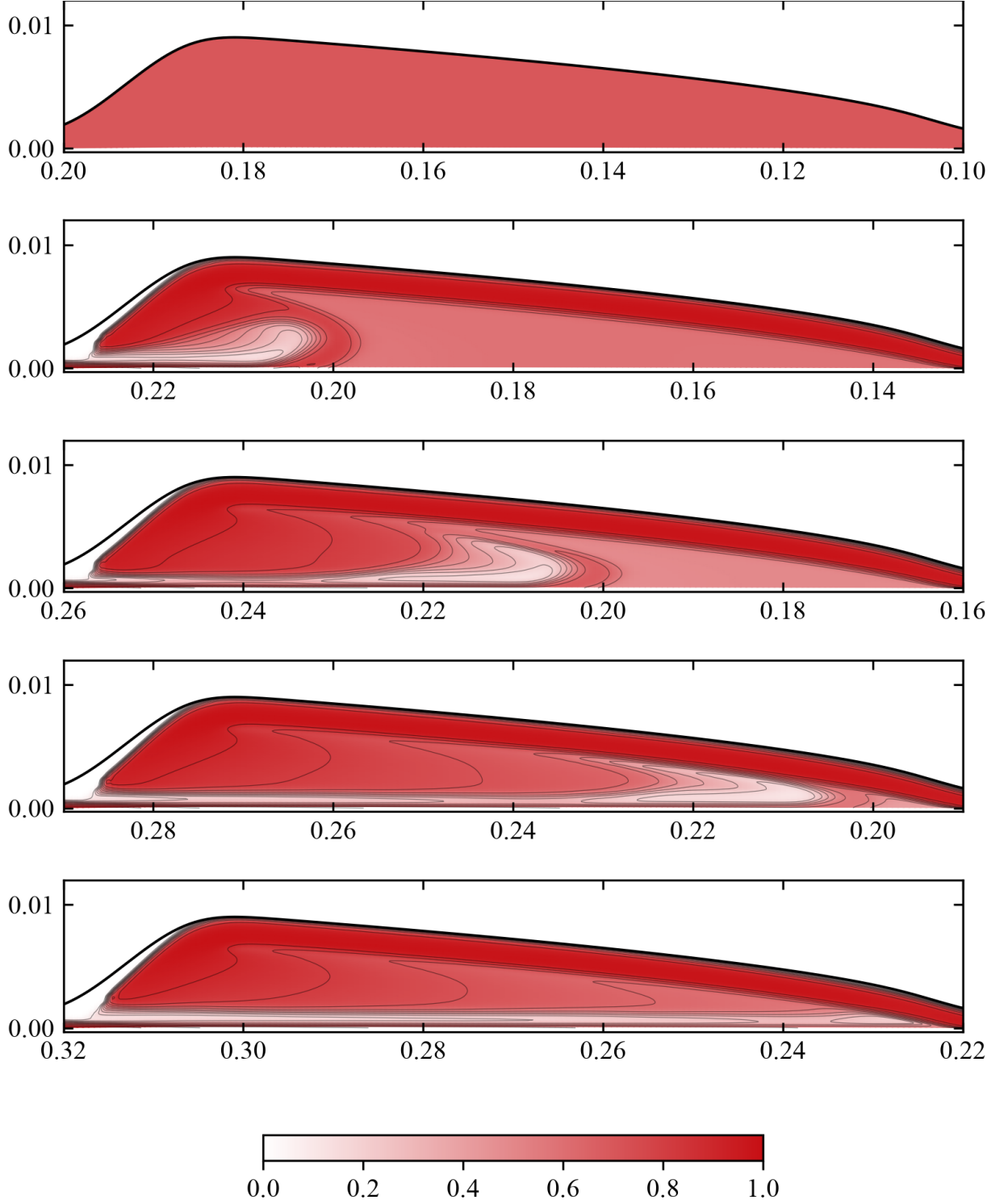


Figure 1: **Compositional fan plot** for the run *Slope comparison*,  $\phi_s^0 = 0.70$ , slope  $i_1 = 0.00975$ ,  $t = 2400$ s. Each “leaf” of the fan corresponds to a temporal snapshot of  $\phi_s(x, \eta)$  in the  $\sigma$ -mesh (normalized vertical coordinate vs. horizontal position), superimposed to visualize the temporal evolution of the segregation pattern from the uniform initial condition (innermost leaf) to the final state (outermost leaf).

## Analysis and interpretation

The fan plot allows visualizing the complete temporal history of segregation over the full dune geometry. Three key features emerge:

- **Sorting front propagation velocity:** the angular separation between consecutive leaves quantifies how rapidly the system departs from the uniform initial state. The front advances quickly at first and progressively slows down, indicating that the system approaches (without reaching) a quasi-steady pattern within the simulated time.
- **Spatial modulation by dune geometry:** the effective front thickness varies with  $x$ , reflecting the locally variable bed height  $h(x)$ . Near the crest (thicker bed), the segregation front has more column to traverse and reaches a more developed state; near the stoss toe (thinner bed), the process is less complete.
- **Approach to a traveling wave:** if the late leaves collapse onto each other, the system tends toward a traveling-wave solution in the co-moving frame. The analysis shows that the system still moves by  $\sim 11.5\%$  of the total change between  $t = 2400$  and  $4800$  s (see Appendix A), confirming that the final snapshot is representative but not fully stationary.

The fan thus motivates the subsequent sections: it shows that segregation builds a persistent large-scale pattern, whose vertical structure is characterized in Section 3 and whose physical mechanism is identified in Section 4.

### 3 Averaged Vertical Sorting Profile

While the fan reveals the space–time evolution, the averaged vertical profile  $\langle\phi_s\rangle_x(\eta)$  condenses the outcome into a single diagnostic that characterizes the final compositional state of the deposit.

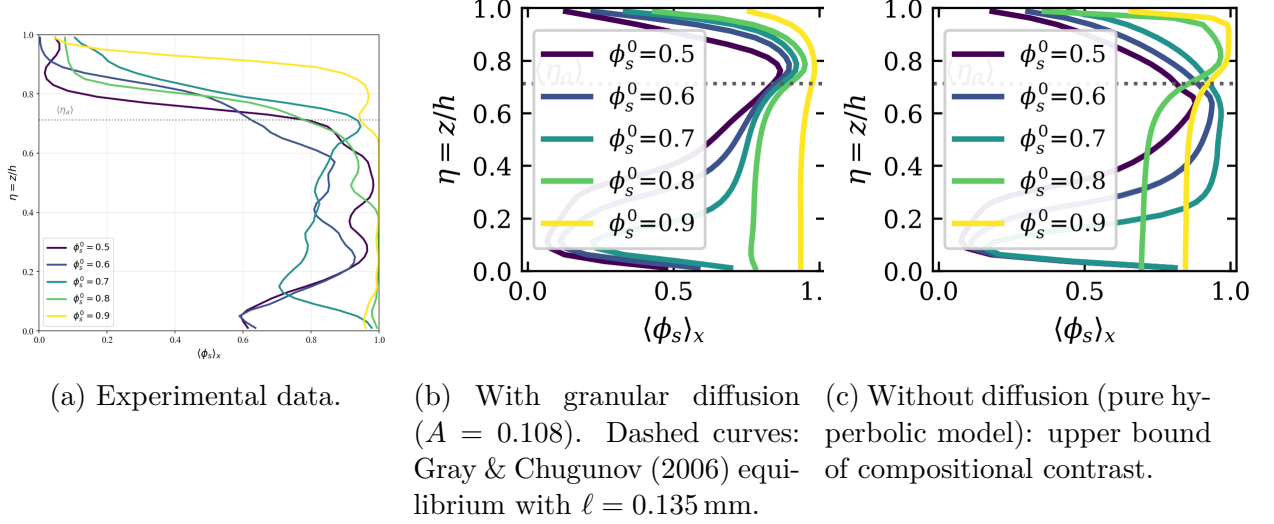


Figure 2: **Vertical sorting profile averaged in  $x$  over the dune body.** Panel (a) shows the experimental data. Panels (b) and (c) show the simulated profiles ( $t = 2400$  s,  $i = 0.00975$ ) for five initial concentrations  $\phi_s^0 \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$ . The horizontal dotted line marks  $\langle\eta_a\rangle \approx 0.71$  (mean roof of the buried deposit).

### Analysis and interpretation

Figure 2 shows the compositional profile averaged along the dune body for the experiment and both model configurations:

**Panel (a) — experimental data.** The experimental profiles establish the physical baseline, exhibiting the characteristic three-layer pattern (coarse base, fines-rich body, and coarse surface crust) that our numerical models aim to reproduce.

**Panel (b) — with diffusion.** All runs exhibit the same **three-layer** pattern, which is the key structural signature of the segregation process:

1. *Base* ( $\eta \lesssim 0.3$ ): enriched in coarse grains. This is a basal crust formed early, when the initial uniform mixture was first exposed to segregation and the coarse grains quickly settled before being permanently buried.
2. *Body* ( $0.3 \lesssim \eta \lesssim \eta_a$ ): a positive gradient of fines upward, direct evidence of active vertical segregation within the mobile active layer.
3. *Surface* ( $\eta \gtrsim \eta_a$ ): a return to low  $\phi_s$  values, forming a coarse surface crust. Segregation continually expels coarse grains upwards, which are re-entrained by the flow before burial.

### Physical definition of the active layer base ( $\eta_a$ )

The dimensionless active layer base  $\eta_a$  separates the permanently buried deposit from the actively shearing mobile layer. It is defined locally as  $\eta_a(x) = 1 - \delta_a/h(x)$ , where  $\delta_a$  is the physical thickness of the active layer (fixed at 1.5 mm in the model) and  $h(x)$  is the local dune height. Because the dune height  $h(x)$  varies spatially, the horizontal dotted line in Figure 2 represents the spatial average over the dune body,  $\langle \eta_a \rangle = \langle 1 - \delta_a/h(x) \rangle \approx 0.713$ . This means that, on average, the upper 28.7% of the column is actively flowing, while the bottom 71.3% constitutes the frozen, buried deposit. The dashed curves represent the theoretical equilibrium profile predicted by Gray & Chugunov (2006) for a steady segregation–diffusion balance. In its dimensional form, the local volumetric fraction of fines  $\phi_s(z)$  at a height  $z$  from the bed is expressed as:

$$\phi_s(z) = \frac{1}{1 + A \exp\left(\frac{f_{sl}z}{D_{sl}}\right)} \quad (2)$$

where  $f_{sl}$  is the maximum segregation velocity and  $D_{sl}$  is the granular diffusion coefficient. Expressed in dimensionless form, scaling the height with the total flow thickness  $h$  via the variable  $\eta = z/h$ , the equation becomes:

$$\phi_s(\eta) = \frac{1}{1 + A \exp(Pe \cdot \eta)} \quad (3)$$

where  $Pe = \frac{f_{sl}h}{D_{sl}}$  is the segregation Péclet number (quantifying the relative strength of segregation versus diffusion), and  $A$  is an integration constant determined by mass conservation.

The function of these curves is to demonstrate that the system relaxes *partially* towards this diffusive steady state within the mobile active layer, but the rapid burial of sediment downwards interrupts the segregation process before the equilibrium can be fully reached.

**Panel (c) — without diffusion.** The profiles are more pronounced: the interfaces are sharper because without  $D_{sl}$  no mechanism broadens them. This panel establishes the **upper limit** of the compositional contrast achievable in the simulated regime, and serves as a reference to isolate the diffusion effect (Section 5). The three-layer pattern is identical in structure, confirming that it is driven by the advective segregation flux rather than by diffusion.

The gradation index  $\langle \phi_s \rangle_{\text{top}} - \langle \phi_s \rangle_{\text{bottom}}$  is positive for all concentrations ( $\phi_s^0 \leq 0.7$ ), confirming that the deposit **fines upwards**. It decreases as  $\phi_s^0$  increases, driven by the non-linearity  $\phi_s(1 - \phi_s)$  of the segregation flux. Granular diffusion reduces the index by  $\approx 20\%$  without altering the qualitative trend (Table 1).

|                   | $\phi_s^0 = 0.5$ | $\phi_s^0 = 0.6$ | $\phi_s^0 = 0.7$ | $\phi_s^0 = 0.8$ | $\phi_s^0 = 0.9$ |
|-------------------|------------------|------------------|------------------|------------------|------------------|
| With diffusion    | +0.48            | +0.44            | +0.27            | +0.04            | +0.01            |
| Without diffusion | +0.53            | +0.51            | +0.35            | +0.05            | +0.02            |

Table 1: Gradation index  $\langle \phi_s \rangle_{\text{top}} - \langle \phi_s \rangle_{\text{bottom}}$  for the five initial concentrations.

## 4 Velocity Field in the Co-moving Frame and Trapped Coarse Zone

The three-layer pattern identified in Section 3 raises the central mechanistic question: *what keeps the coarse grains trapped in the interior of the dune?* This section addresses it by inspecting the flow field in the reference frame co-moving with the dune.

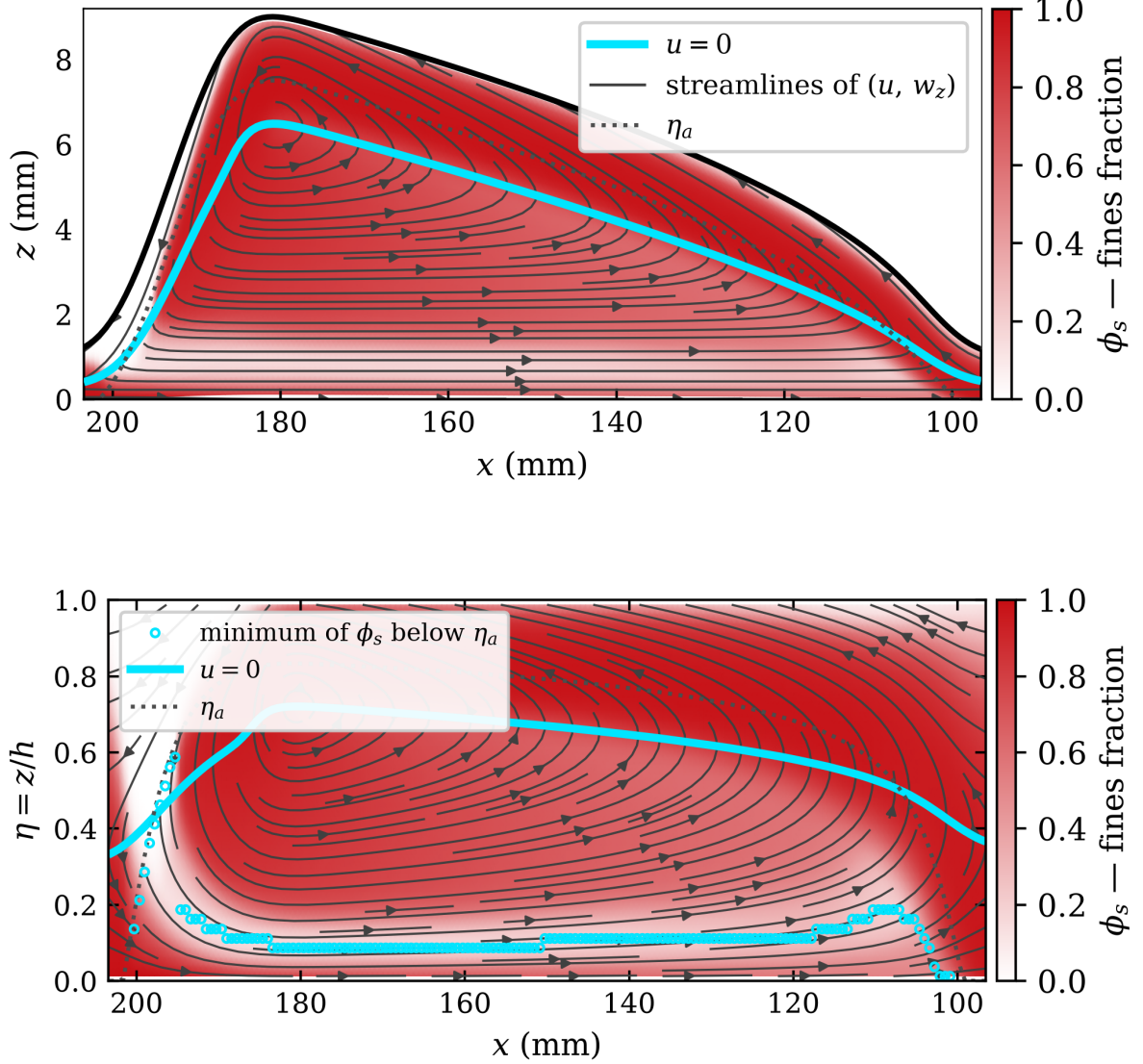


Figure 3: **The trapped coarse zone and the recirculation cell** ( $\phi_s^0 = 0.70$ , with diffusion,  $t = 2400$  s). **(a)** Physical coordinates:  $\phi_s(x, z)$ , streamlines of  $(u, w_z)$  in the co-moving frame with the dune, and isoline  $u = 0$  (cyan). **(b)**  $\sigma$ -mesh (rectangular): the same field in coordinates  $(x, \eta)$ . The cyan circles mark the minimum of  $\phi_s$  below  $\eta_a$  (coarse-enriched zone).

### Analysis and interpretation

**Panel (a) — physical coordinates.** The isoline  $u = 0$  (cyan) in the co-moving frame delimits the material that *overtakes* the migrating dune from material that *falls behind*

and gets permanently buried. A naive hypothesis would place the coarse enrichment right at or below this isoline. However, the figure shows that the coarse-rich zone (light colors in the interior) does not coincide spatially with the  $u = 0$  isoline: the “kinematic stagnation point” hypothesis is visually refuted.

**Panel (b) —  $\sigma$ -mesh.** The  $\sigma$  (terrain-following) coordinate system reveals the full structure of the recirculation. Quantitative analysis confirms the spatial mismatch:

| Magnitude                                    | Value              |
|----------------------------------------------|--------------------|
| $\eta(u = 0) - \eta(\text{coarse})$ , median | +0.527             |
| Range                                        | $[-0.098, +0.633]$ |
| Equivalent in mm                             | +3.27 mm           |
| Evaluated columns                            | 159                |
| Pearson correlation                          | −0.54              |

Table 2: Separation between the  $u = 0$  isoline and the coarse core.

The **anti-correlation** (Pearson = −0.54) is the most important result: where the  $u = 0$  isoline rises, the coarse core descends. The true trapping mechanism is the **closed recirculation cell** visible in both panels, centered at  $x \approx 180$  mm,  $\eta \approx 0.55$ . Sediment that enters the cell cannot exit it within the simulation time, remaining indefinitely confined in the interior of the dune. This is analogous to the particle recirculation described by Thornton & Gray (2008) for size-segregation in dense granular flows, *not* to a simple stagnation point.

## 5 Controlled Comparison: Effect of Granular Diffusion

Having established the baseline structure (Sec. 3) and its physical origin (Sec. 4), we now isolate the contribution of granular diffusion by directly comparing the with-diffusion and without-diffusion runs.

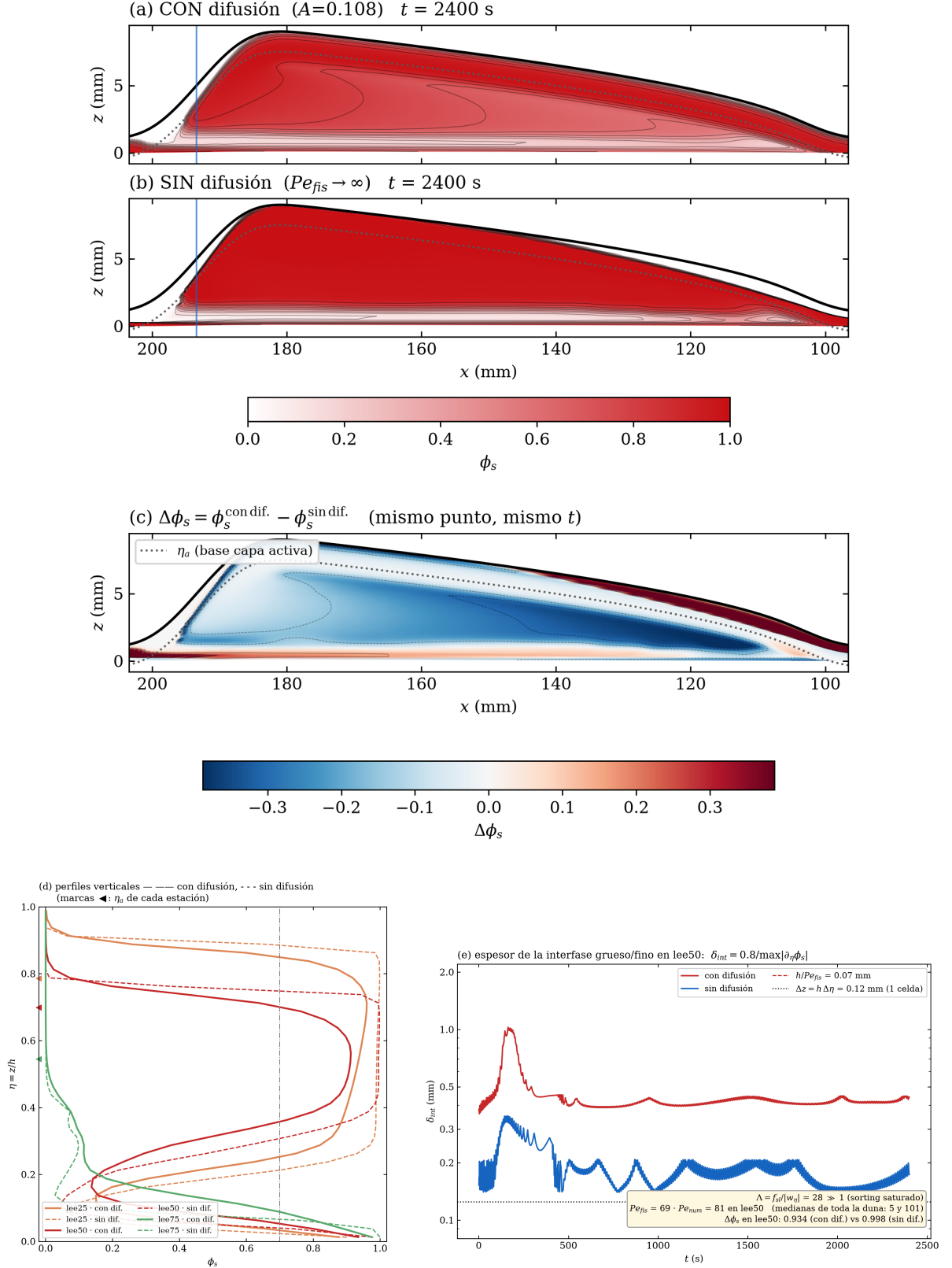


Figure 4: **Comparison of models with and without granular diffusion** ( $\phi_s^0 = 0.70$ ,  $t = 2400$  s). **Top (panels a & b)**: The  $\phi_s$  fields with diffusion (a) and without diffusion (b), displayed on the same colorscale. **Middle (panel c)**: Pointwise difference  $\Delta\phi_s = \phi_s^{\text{diff}} - \phi_s^{\text{no diff}}$ ; red indicates zones where diffusion adds more fines, blue where it adds more coarse. **Bottom (panels d & e)**: Vertical profiles  $\langle \phi_s \rangle_x(\eta)$  comparing both models (d) and the temporal evolution of the coarse/fine interface thickness  $\delta_{int}$  (e).

## Analysis and interpretation

Figure 4 isolates the effect of the granular diffusion term by a direct controlled comparison.

- **Panels (a) and (b) — global structure:** The overall layout of the recirculation cell, the three-layer stratification, and the coarse core position are essentially identical in both models. This demonstrates that the *topology* of the segregation pattern is dominated by the advective segregation flux, not by diffusion. Diffusion is a perturbative, not a structurally dominant, process.
- **Panel (c) — pointwise difference:** The difference  $\Delta\phi_s$  is concentrated at the compositional interfaces (boundaries of the layers identified in Sec. 3), where diffusion broadens the transitions. Key metrics:
  - **RMS** of  $\Delta\phi_s$ : 0.277
  - **Max**  $|\Delta\phi_s|$ : 0.944
  - **Color scale saturated at**  $\pm 0.39$
  - $|\Delta\phi_s| > 0.05$  in 71% of the dune volume.
- **Panel (d) — averaged profiles:** Diffusion reduces the gradation index by  $\approx 20\%$  (consistent with Table 1) but does not change the sign or the qualitative shape of the profile.
- **Panel (e) — interface thickness:** This panel provides the most direct evidence for the physical role of diffusion. The thickness  $\delta_{\text{int}}$  is defined based on the maximum concentration gradient as:

$$\delta_{\text{int}} = h \frac{0.8}{\max \left| \frac{\partial \phi_s}{\partial \eta} \right|} \quad (4)$$

This formula represents the spatial width of a linear ramp spanning 80% of the composition change (from  $\phi_s = 0.1$  to  $0.9$ ). Evaluating the 10%–90% interval is a standard physical convention that isolates the primary transition core, avoiding the infinitely long asymptotic tails characteristic of diffusive profiles. Without diffusion (blue curve), the interface collapses toward the numerical mesh limit ( $\approx 0.19$  mm,  $\sim 1.5$  cells). With diffusion (red curve), the thickness stabilizes smoothly above the numerical floor at  $\approx 0.45$  mm, consistent with the static equilibrium  $\ell \approx 0.07$  mm modified by advective stretching during burial. This validates that the modeled broadening is a real physical effect, not a grid artifact.

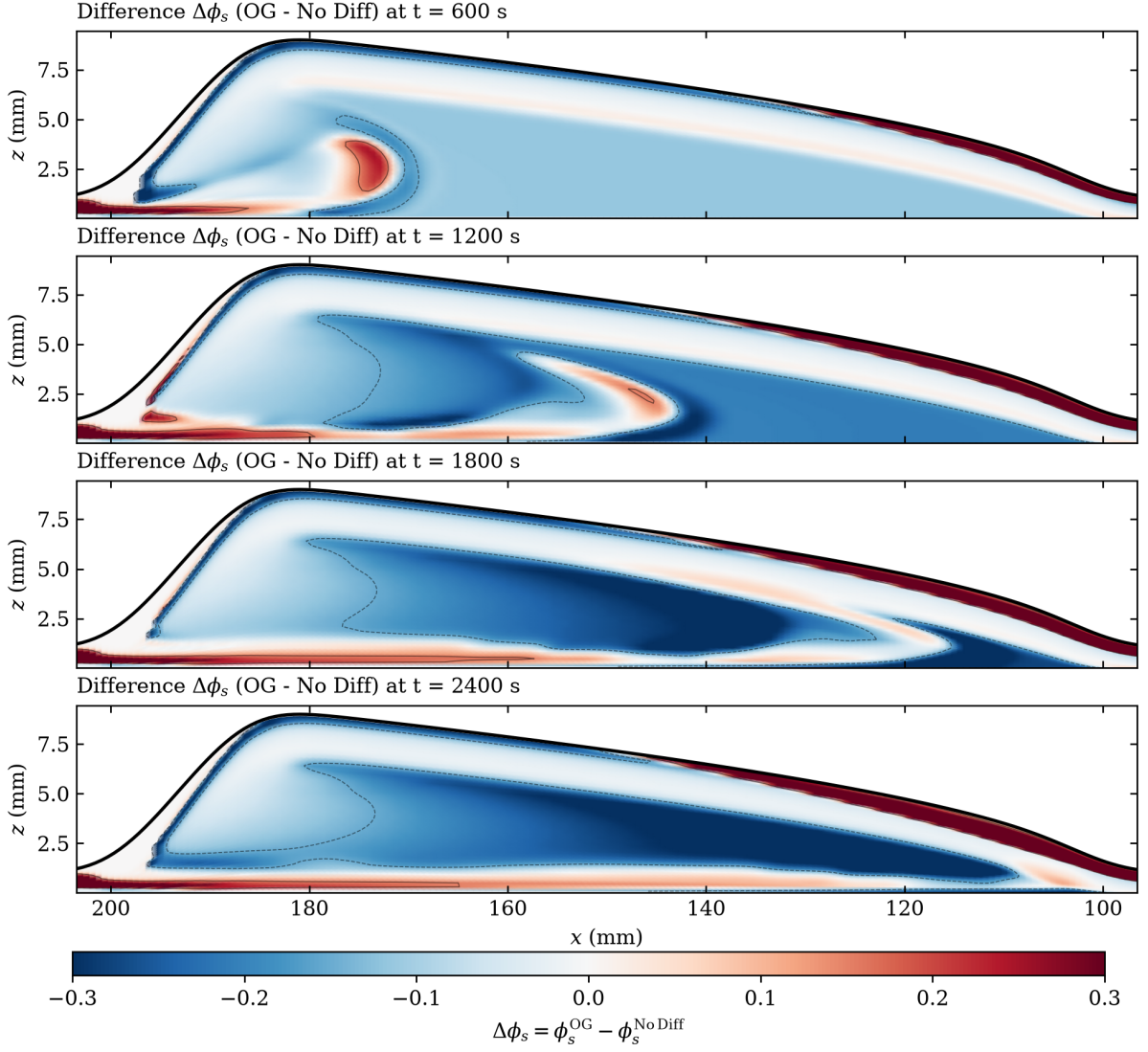
### Physical interpretation: the sorting front

The **sorting front** is defined as the boundary below which the segregation process has substantially modified the initial uniform mixture ( $|\phi_s - \phi_s^0| > 0.05$ ). Its dynamics explain why the system reaches the final state it does:

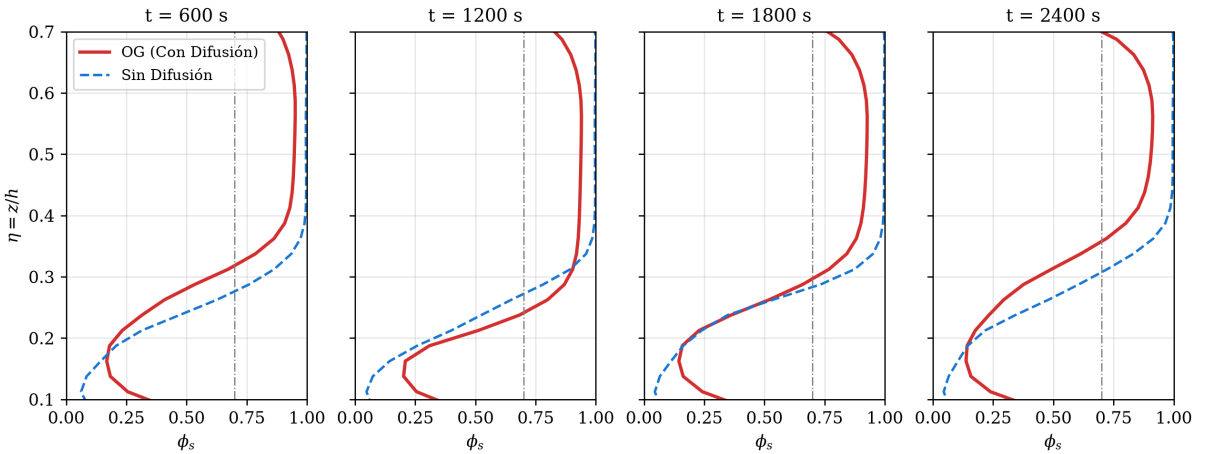
1. **Rapid descent** ( $t \lesssim 300$  s): The front descends from the surface (active layer) into the dune interior faster than the burial rate, because the downward flux of fines by gravity augments the passive burial velocity.
2. **Physical arrest** ( $t \approx 300$  s): The front reaches the impermeable base ( $\eta = 0$ ). From this point, the entire column has been visited by segregation and no virgin sediment remains. The simulation then runs to  $t = 2400$  s to capture the slower spatial reorganization that builds the recirculation cell (Sec. 4).

## Temporal Evolution of the Interface Difference

To further illustrate how diffusion broadens the interfaces dynamically, we track the point-wise difference  $\Delta\phi_s$  over time. Figure 5 shows this temporal evolution across the dune, alongside a detailed zoom of the vertical profile at the center of the avalanche face.



(a) Spatial maps of the difference  $\Delta\phi_s$  at successive times.



(b) Zoomed vertical profiles of  $\phi_s$  at the 1ee50 station, highlighting the interface thickness.

Figure 5: **Temporal evolution of the difference between the models with and without diffusion.** The maps (a) track how the broadened interfaces propagate downwards over time, while the zoomed vertical profiles (b) contrast the smooth physical transition of the OG model (red) against the abrupt step of the hyperbolic model (blue dashed line).

Table 3 quantifies this evolution at the `1ee50` station, extracted directly from the data of Figure 4(e). The purely hyperbolic model (without diffusion) collapses quickly towards the numerical resolution limit (oscillating around  $\approx 0.17\text{--}0.19\text{ mm}$ , which corresponds to  $\sim 1.5$  grid cells), whereas the model with granular diffusion stabilizes the interface physically, maintaining a thickness systematically above  $0.4\text{ mm}$  during the late stages of segregation.

| Time (s) | $\delta_{\text{int}}$ With Diffusion (mm) | $\delta_{\text{int}}$ Without Diffusion (mm) |
|----------|-------------------------------------------|----------------------------------------------|
| 100      | 0.512                                     | 0.197                                        |
| 300      | 0.476                                     | 0.259                                        |
| 600      | 0.396                                     | 0.185                                        |
| 900      | 0.419                                     | 0.175                                        |
| 1200     | 0.403                                     | 0.177                                        |
| 1500     | 0.439                                     | 0.180                                        |
| 1800     | 0.405                                     | 0.171                                        |
| 2100     | 0.413                                     | 0.142                                        |
| 2400     | 0.441                                     | 0.173                                        |

Table 3: Temporal evolution of the interface thickness  $\delta_{\text{int}}$  measured at the `1ee50` station.

## 6 Temporal Evolution of $\phi_s$ : Kinetics of Vertical Sorting

The fan (Sec. 2) and the final averaged profile (Sec. 3) frame the beginning and end states. Here we focus on the *path* between them: how the vertically averaged profile  $\langle\phi_s\rangle_x(\eta)$  evolves in time, and how diffusion alters that path.

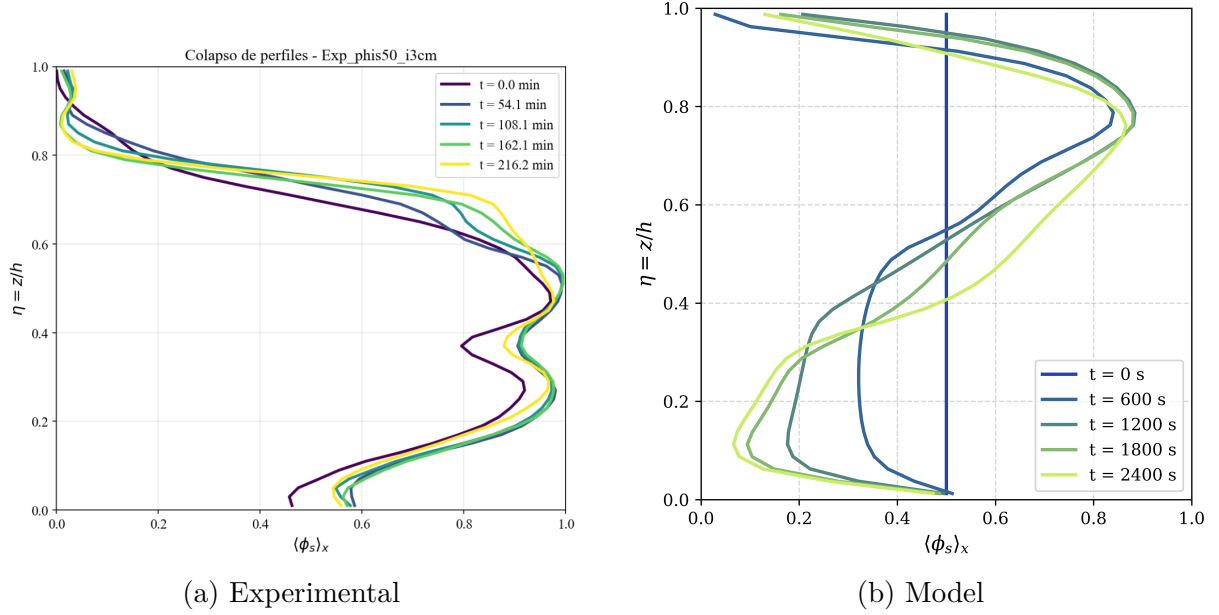


Figure 6: **Temporal evolution of the vertical sorting for  $\phi_s^0 = 0.50$ .** Left: Experimental data. Right: Model prediction.

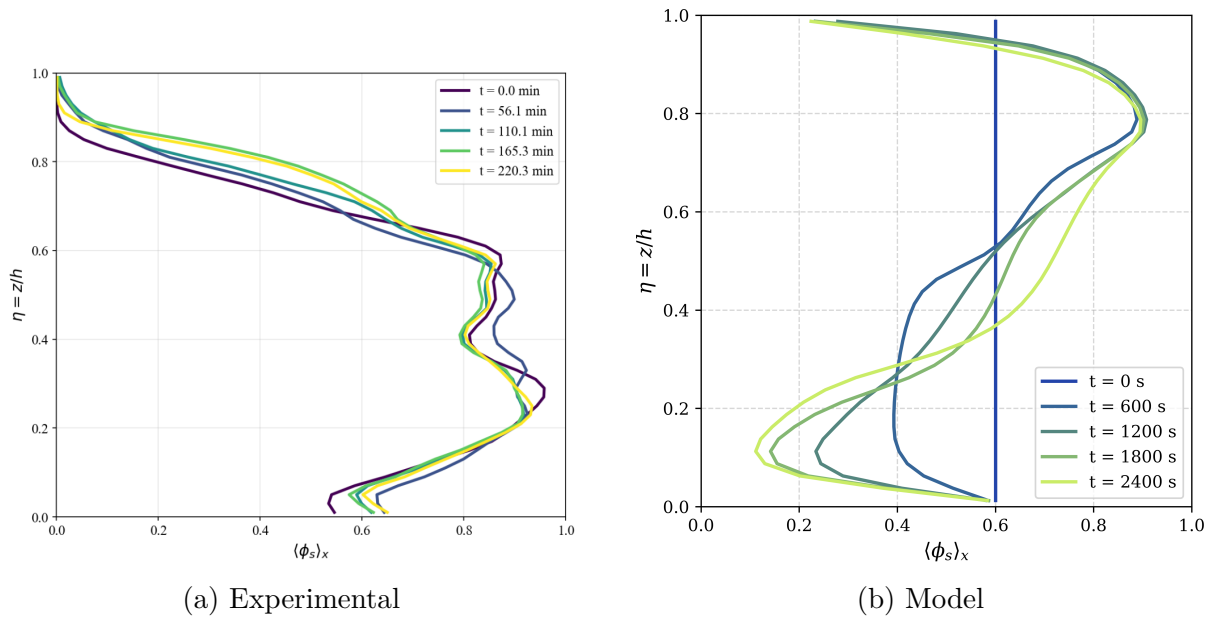


Figure 7: **Temporal evolution of the vertical sorting for  $\phi_s^0 = 0.60$ .** Left: Experimental data. Right: Model prediction.

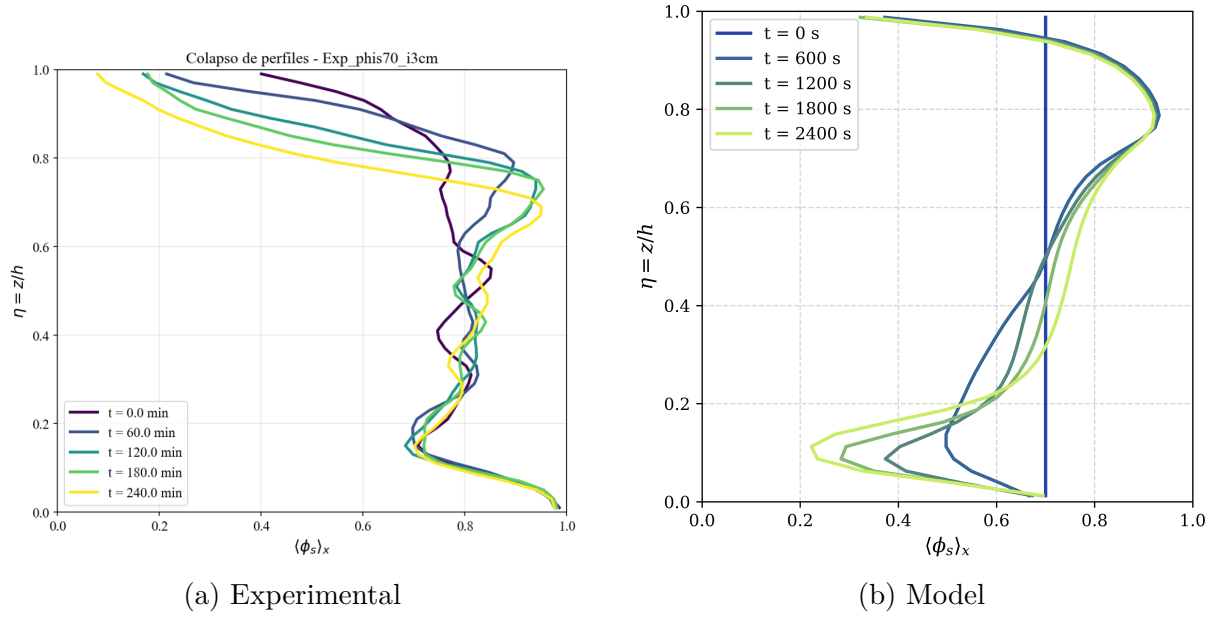


Figure 8: **Temporal evolution of the vertical sorting for  $\phi_s^0 = 0.70$ .** Left: Experimental data. Right: Model prediction.

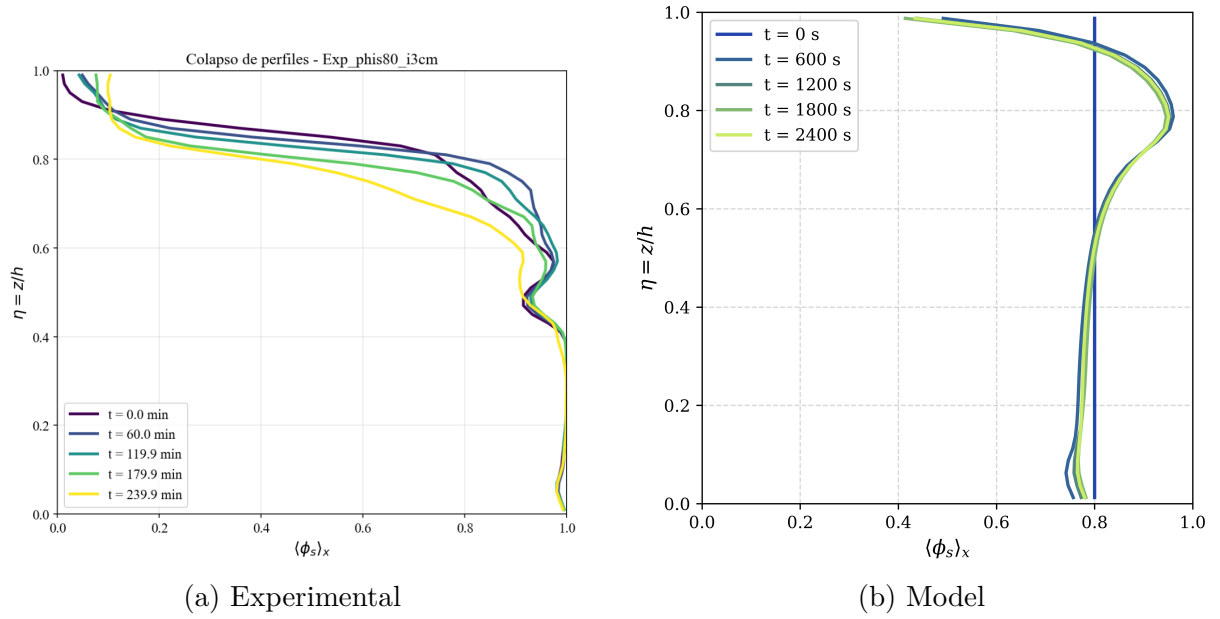


Figure 9: **Temporal evolution of the vertical sorting for  $\phi_s^0 = 0.80$ .** Left: Experimental data. Right: Model prediction.

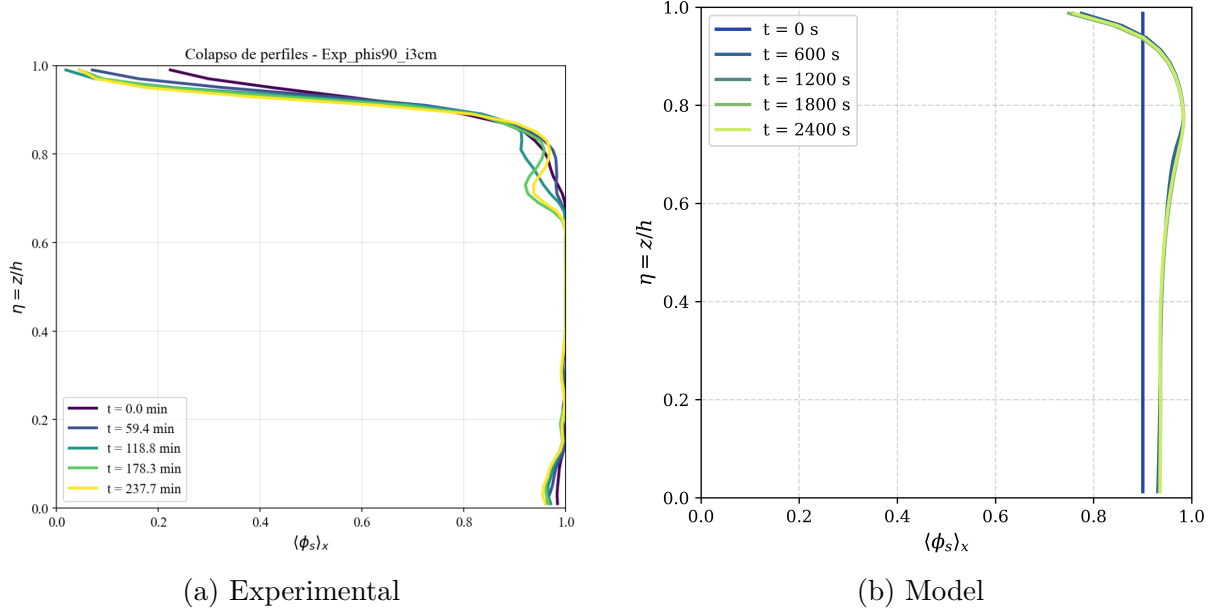


Figure 10: **Temporal evolution of the vertical sorting for  $\phi_s^0 = 0.90$ .** Left: Experimental data. Right: Model prediction.

## Analysis and interpretation

The side-by-side comparison of experimental data and model predictions across all initial concentrations reveals the **kinetics** of the segregation process. Both experimental and numerical profiles demonstrate three kinetic stages:

1. *Induction stage* ( $t \lesssim 500$  s): The profile is nearly unchanged from the initial uniform condition (dotted line). The velocity field is being established and the segregation flux has not yet had time to produce a visible effect on the column-averaged profile.
2. *Active transient* ( $500 \lesssim t \lesssim 1800$  s): The sorting front (Sec. 5) descends rapidly. Fines accumulate in the middle layers of the column while coarse grains are expelled upward, building the characteristic three-layer structure.
3. *Deceleration* ( $t \gtrsim 2000$  s): The profile changes progressively slower as the system approaches the diffusive equilibrium within the active layer. The process does not fully stop within the simulation time (see Appendix A).

The uniform initial condition (dotted line) serves as the reference for the *total change* from start to finish, which is the appropriate denominator when normalizing convergence metrics.

## 7 Equilibrium Length $\ell$ and Local Péclet Fields

The heatmaps of  $\ell$  and  $Pe$  provide the quantitative framework that explains *why* the advective segregation flux dominates and what role the lithostatic pressure plays in suppressing segregation at depth.

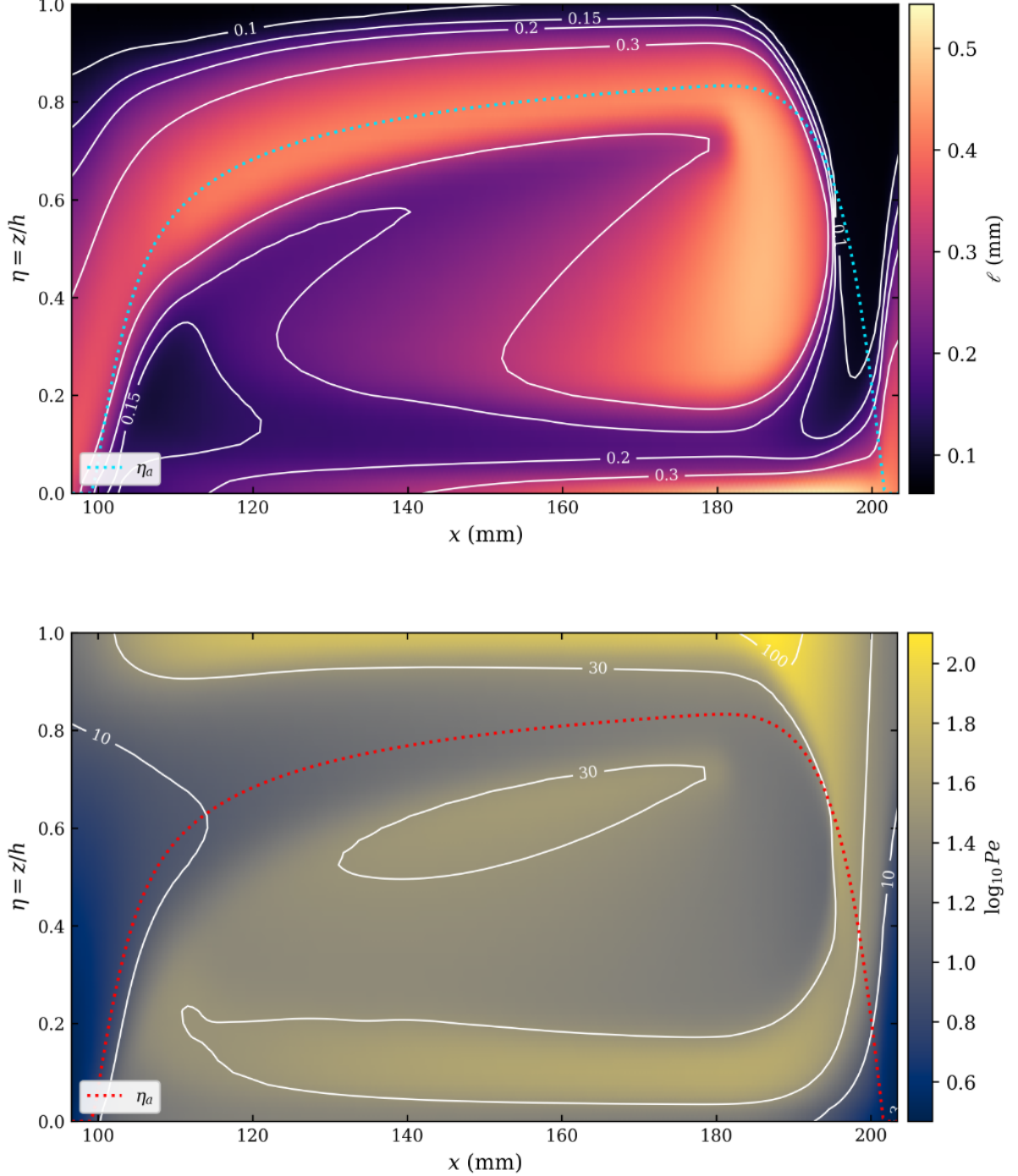


Figure 11: **Spatial maps of  $\ell(x, \eta)$  and  $Pe(x, \eta)$  for  $\phi_s^0 = 0.70$ , with diffusion,  $t = 2400$  s.** **Top:** map of the equilibrium length  $\ell$  (mm) with white isolines. The increase of  $\ell$  with depth reflects the growth of lithostatic pressure. **Bottom:** map of  $\log_{10} Pe$  with white isolines.  $Pe \gg 1$  throughout the dune confirms that segregation advection dominates over diffusion at all spatial locations.

## Analysis and interpretation

**Map of  $\ell$ .** The equilibrium length  $\ell = D_{sl}/f_{sl}$  grows with depth because the lithostatic pressure  $p$  in the numerator of Eq. (1) increases towards the base. Typical values range from  $\ell \approx 0.10$  mm at the surface to  $\ell \approx 0.30$  mm at the base (a factor of  $\approx 8.7$ ). This implies that compositional interfaces are inherently more diffuse at depth, independent of whether an explicit diffusion term is included. The lithostatic pressure thus acts as a *natural depth-dependent regularization* of the hyperbolic problem.

**Map of  $Pe$ .** With  $L = h$  (the full bed thickness as the reference length):

- $Pe$  increases towards the surface (where pressure is low and  $\ell$  is small) and towards the crest (where  $h$  is largest).
- $Pe \gg 1$  throughout the entire dune: at the scale of the full column, *segregation always dominates diffusion*.

Using the physically tighter scale  $L = \delta_a$  (active layer thickness) reduces  $Pe$  considerably but does not invert the conclusion at the reference station (lee 50%,  $h = 4.98$  mm):

|                          | Surface | Base |
|--------------------------|---------|------|
| $Pe$ with $L = h$        | 79.8    | 9.2  |
| $Pe$ with $L = \delta_a$ | 24.0    | 2.8  |

Table 4: Péclet number at the lee 50% station.

$Pe > 1$  everywhere in the column even with  $L = \delta_a$ , confirming that segregation advection is dominant. The narrowest margin is at the base ( $Pe \approx 2.8$ ), where diffusion provides the strongest relative broadening —consistent with the wider interfaces seen at depth in the  $\ell$  map.

**Note on the regularization floor.** The comparison with and without  $p_{\text{floor}}$  shows that  $p_{\text{floor}} \approx p_{\text{lit}}^{\text{max}}$ : a fraction of the depth-dependent suppression of segregation is numerical regularization rather than genuine physical lithostatic pressure. This must be declared explicitly in the paper.

## 8 Summary of the Main Results

| Figure           | Main result                                                                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fig. 1 (§2)      | Global space–time picture. The sorting front propagates rapidly then slows; the pattern is spatially modulated by dune geometry and approaches (but does not reach) a traveling wave.                                       |
| Fig. 2 (§3)      | Three universal compositional layers (coarse base, fines body, coarse surface crust). Diffusion reduces the gradation index by $\approx 20\%$ . The index collapses for $\phi_s^0 \rightarrow 1$ .                          |
| Fig. 3 (§4)      | The trapping of coarse grains is driven by a closed recirculation cell in the co-moving frame, <i>not</i> by the kinematic stagnation point. Pearson anti-correlation = $-0.54$ .                                           |
| Fig. 4 (§5)      | Diffusion broadens interfaces and reduces contrasts by $\approx 20\%$ but does not alter the topology of the recirculation cell or the three-layer structure. Interface thickness validated above the numerical mesh floor. |
| Fig. 5 (§5)      | The dynamic difference maps and zoom profiles reveal how the interface broadening induced by diffusion evolves and deepens continuously over time.                                                                          |
| Figs. 6–10 (§6)  | Three kinetic stages: induction, active transient, deceleration. The state at $t = 2400$ s is representative but not fully converged (see Appendix A).                                                                      |
| Fig. 11 (§7)     | $Pe \gg 1$ throughout the dune with $L = h$ ; with $L = \delta_a$ the margin narrows at the base. Lithostatic pressure naturally suppresses segregation in depth.                                                           |
| Fig. 12 (App. A) | The system moves by $\sim 11.5\%$ of the total change between $t = 2400$ and $4800$ s: not fully converged.                                                                                                                 |

Table 5: Synthesis of the key results by figure.

### General conclusion

The set of figures demonstrates that **bidisperse sedimentary segregation in hydraulic dunes** produces a robust and repeatable compositional structure. This structure is:

1. **Qualitatively insensitive** to the hydraulic slope in the range  $i \in [0.005, 0.014]$ : the slope controls the speed of unmixing, not the sharpness of the interfaces ( $\ell$  and  $Pe$  do not depend on  $i$  because  $\dot{\gamma}$  cancels in  $D_{sl}/f_{sl}$ ).
2. **Quantitatively reduced** by granular diffusion by  $\approx 20\%$ , with no qualitative change in the trend.

3. **Generated by advection**, not by kinematic stagnation: the trapping mechanism is closed recirculation in the co-moving frame (Thornton & Gray 2008), with an anti-correlation  $r = -0.54$  with respect to the position of  $u = 0$ .
4. **Not completely steady** at  $t = 2400$  s: the field continues to evolve, although most of the transient has elapsed.

# A Temporal Convergence Analysis

This appendix documents the degree to which the simulations have converged at the reporting time  $t = 2400$  s. The analysis uses an extended run at  $t = 4800$  s as a reference.

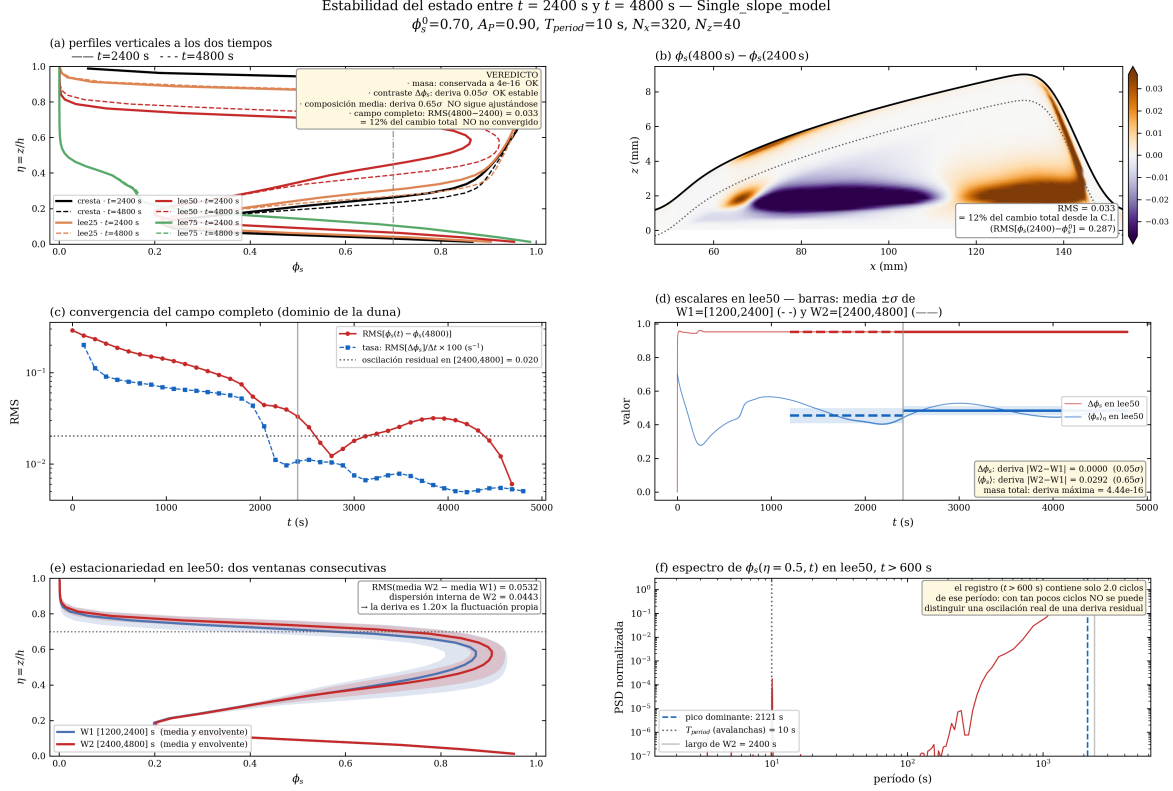


Figure 12: **Temporal stability analysis**: comparison of the compositional field between  $t = 2400$  s and  $t = 4800$  s. The only extended run uses the *Single slope model*, mesh  $N_x = 320$ ,  $\phi_s^0 = 0.70$ , with diffusion. The main panel shows the relative  $L_2$  norm of the time-doubling test; the secondary panels show alternative metrics (geometric tracking of the coarse core and instantaneous rate of change).

## Time-doubling test

For each instant  $t$ , the field is compared against its doubled time:

$$\epsilon(t) = \frac{\|\phi_s(t) - \phi_s(2t)\|_2}{\|\phi_s(2t)\|_2}$$

| Magnitude                                        | Value            |
|--------------------------------------------------|------------------|
| Relative $L_2$ norm at $t = 2400$ s              | 0.046            |
| Minimum value (at $t \approx 2040$ s)            | 0.025            |
| $\ \phi(4800) - \phi(0)\ $ (total change)        | 23.16            |
| $\ \phi(2400) - \phi(4800)\ $                    | 2.65             |
| Fraction of total change between 2400 and 4800 s | $\approx 11.5\%$ |

Table 6: Convergence metrics of the run extended to 4800 s.

**Correct interpretation.** The denominator  $\|\phi_s(2t)\|_2$  is dominated by the mean composition ( $\langle\phi_s\rangle \approx 0.667$ ), not by the structural variation. When the residual is instead normalized by the *total change from the initial condition*, the system is found to have moved  $\sim 11.5\%$  between  $t = 2400$  and  $4800$  s. The state at  $t = 2400$  s is therefore representative of the late-transient regime but is not an equilibrium: reporting it as “steady” requires an explicit caveat.

**Alternative metrics.** The geometric tracking of the coarse core (mean  $\eta$  position of the  $\phi_s$  minimum below  $\eta_a$ ) confirms that the core continues to descend slowly beyond  $t = 2400$  s. The instantaneous rate of change  $d\|\phi_s\|/dt$  decays monotonically after  $t \approx 2040$  s, confirming that the system is decelerating but not halting within the simulated window.